

1. Administrative & Study Identification

Full Study Title: Efficacy and Safety of Pembrolizumab (MK-3475) Plus Lenvatinib (E7080/MK-7902) in Previously Treated Participants With Select Solid Tumors (MK-7902-005/E7080-G000-224/LEAP-005) **Official Title:** A Multicenter, Open-label Phase 2 Study of Lenvatinib (E7080/MK-7902) Plus Pembrolizumab (MK-3475) in Previously Treated Subjects With Selected Solid Tumors (LEAP-005) **Short Title/Acronym:** LEAP-005 **Protocol Number:** MK-7902-005 / E7080-G000-224 **NCT Number:** NCT03797326 **PostingID:** 452829403385667383 **Sponsor / Company:** Merck **Investigational Product:** Lenvatinib (anti-angiogenic multikinase inhibitor) plus Pembrolizumab (anti-PD-1 monoclonal antibody) **Phase of Development:** PHASE2 (Phase ID: PHASE2; Criteria Type: PHASE_REQUIREMENT; Criteria Text: Trial must be in PHASE2) **Trial Status:** COMPLETED **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the safety and efficacy of combination therapy with pembrolizumab (MK-3475) and lenvatinib (E7080/MK-7902) in participants with select previously treated solid tumors.

2.2 Background & Rationale To address the critical unmet medical need for novel and effective treatments in patients with advanced, metastatic, or unresectable solid tumors that have progressed on or after prior standard systemic therapies. **Disease Areas Evaluated:** Advanced Solid Tumors, Triple Negative Breast Cancer, Ovarian Cancer, Gastric Cancer, Colorectal Cancer, Glioblastoma, Biliary Tract Cancers, Pancreatic Cancer.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Multi-cohort) **Blinding:** Open-label **Randomization:** N/A (Non-randomized)

3.2 Planned Enrollment & Duration Targeted Enrollment: 611 participants **Number of Sites:** Global (Over 90 sites across multiple countries) **Estimated Study Start:** March 2019 **Trial Status Completion:** COMPLETED (Status Name: COMPLETED; Status Description: Study has been completed)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 within 3 days of study treatment initiation. 2. Histologically or cytologically-documented, advanced (metastatic and/or unresectable) solid tumor that is incurable and for which prior standard systemic therapy has failed. Cohort-specific prior line requirements: - **TNBC**: 1-2 prior lines, LDH $<2.0 \times$ ULN, and known ER/PR/HER2 status. - **Ovarian**: Primary ovarian cancer; 3 prior lines. - **Gastric**: 2 prior lines (includes GEJ; excludes squamous cell carcinoma). - **CRC**: 2 prior lines. - **GBM**: Failed initial systemic therapy for newly diagnosed GBM; histologically confirmed WHO Grade IV; neurologically stable; locally determined MGMT analysis. - **BTC**: 1 prior line; Child-Pugh Score Class A (5-6). - **Pancreatic**: Pathologically confirmed pancreatic ductal adenocarcinoma, metastatic at enrollment; 1-2 prior lines (prior therapy with at least 1 platinum- or gemcitabine-containing regimen). 3. Measurable disease per RECIST 1.1 (or RANO criteria for the GBM cohort) confirmed by BICR. 4. Provided a PD-L1 evaluable archival tumor tissue sample or newly obtained biopsy. 5. Adequate organ function and agreement to use approved contraception.

4.2 Exclusion Criteria 1. Gastrointestinal malabsorption or any condition affecting absorption of lenvatinib. 2. Present or progressive accumulation of pleural, ascitic, or pericardial fluid requiring drainage or diuretic drugs within 2 weeks prior to enrollment (except ovarian cancer cohort). 3. Radiographic evidence of encasement/invasion of a major blood vessel, intratumoral cavitation, or clinically significant hemoptysis/tumor bleeding within 2 weeks. Specifically, portal vein invasion (Vp4), inferior vena cava, or cardiac involvement based on imaging in the BTC cohort. 4. Active central nervous system metastases, carcinomatous meningitis, or active autoimmune disease requiring systemic treatment within the past 2 years. 5. Significant cardiovascular impairment within 12 months (e.g., NYHA Class II+ heart failure, unstable angina, MI, CVA). 6. Prior therapy with lenvatinib, anti-PD-1, anti-PD-L1, or anti-PD-L2 agents. 7. Received a live vaccine within 30 days prior to the first dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Lenvatinib 20 mg administered orally once daily; Pembrolizumab 200 mg administered intravenously every 3 weeks. **Pharmacological Identifiers:** - **Drug Name:** lenvatinib (Drug Preferred Name: lenvatinib) - **Trade Names:** Lenvima - **ATC Code:** L01EX08 - **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/lenvima-epar-product-information_en.pdf - **Preferred UMLS Name:** Pembrolizumab - **Lenvatinib Duration of Treatment:** Up to 35 cycles or until

disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care. **Prohibited:** Live virus vaccines, biologic response modifiers within 4 weeks, chronic systemic immunosuppressive therapy (e.g., >10 mg/day prednisone equivalent), and prior CTLA-4, OX 40, or CD137 inhibitors.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Baseline Imaging, PD-L1 Assessment
Baseline	Day 1	Enrollment, First Dose
Treatment	Every 3 Weeks	Pembrolizumab infusion, safety labs, Lenvatinib dispensation
Imaging/Interim	Every 9 Weeks	Efficacy assessment via CT/MRI
End of Study	Follow-up	Safety follow-up 30 days post-last dose (120 days post-pembrolizumab for contraception)

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- Objective Response Rate (ORR) Per RECIST 1.1 by Investigator Assessment:** Defined as the percentage of participants with a best overall response of Complete Response (CR) or Partial Response (PR) (Cohorts A and B).
- Objective Response Rate (ORR) Per RECIST 1.1 or RANO Criteria (for GBM Only), by BICR:** CR or PR (sum of products of diameters decreased by $\geq 50\%$ from baseline for GBM) evaluated by Blinded Independent Central Review (Cohorts C, D1, D2, E, F, and G).
- Number of Participants With One or More Adverse Events (AEs):** Assessment of any untoward medical occurrence temporally associated with the use of study intervention.

7.2 Secondary Outcomes

- Disease Control Rate (DCR) Per RECIST 1.1 by Investigator Assessment.**
- Disease Control Rate (DCR) Per RECIST 1.1 or RANO Criteria (GBM Only) by BICR.**
- Duration of Response (DOR) Per RECIST 1.1 by Investigator Assessment.**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring evaluating any unfavorable and unintended sign temporally associated with intervention use. **Serious Adverse Events (SAEs):** Reported promptly; follow-up extends comprehensively. **Data Monitoring Committee (DMC):** Evaluates periodic safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All Participants as Treated (APaT) for safety analyses; Full Analysis Set (FAS) for efficacy. **Sample Size Justification:** Targeted Enrollment scaled to 611 evaluating primary response limits across solid tumor cohorts. **Handling of Missing Data:** Censoring rules apply for unrecorded progression elements.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Compliant with Good Clinical Practice (GCP). **Audit Trail:** Utilizing compliant EDC systems for securely handling eCRFs and central BICR radiological image processing.

11. Study Identifiers & Governance

Primary ID: {{MK-7902-005}} **Registry Number:** {{NCT03797326}}
PostingID: {{452829403385667383}} **Sponsor/Lead Organization:** {{Merck}} **Company:** {{Merck}} **Study Title:** {{Efficacy and Safety of Pembrolizumab (MK-3475) Plus Lenvatinib (E7080/MK-7902) in Previously Treated Participants With Select Solid Tumors (MK-7902-005/E7080-G000-224/LEAP-005)}} **Database Source Level:** {{5}}

12. Clinical Framework & Taxonomy Mappings

Condition Scope: {{Advanced Solid Tumors, Triple Negative Breast Cancer, Ovarian Cancer, Gastric Cancer, Colorectal Cancer, Glioblastoma, Biliary Tract Cancers, Pancreatic Cancer}}
Anatomical Target Mapping (Example: Biliary Tract): - **Name:** Biliary Tract - **Preferred Name:** Biliary tract structure - **Semantic Type:** Body Part, Organ, or Organ Component - **Definition:** The system that transports bile from the hepatocytes in the liver to the small intestine. It is comprised of the intrahepatic bile ducts, hepatic ducts, common bile duct, cystic duct, and the gallbladder. - **MeSH Code:** D001659 - **SNOMED ID:** 181267003
Optimization Target: {{Objective Response Rate (ORR), Disease Control Rate (DCR), and Safety (AE monitoring)}}

13. Study Procedures & Methodology

Management Pathway: {{Targeted Combination Checkpoint and TKI pathway}} **Education Protocol:** {{Protocol compliance, immune-mediated AE identification, RECIST 1.1/RANO review}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) of imaging}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 35 treatment cycles}} **Onsite Visit**

Cadence: `{{Every 3 weeks (Q3W)}}` **Checklist Review Interval:**

```
{{Tumor imaging every 9 weeks}}
```

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					